

The uniform Taylor tree for $d = 2$

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Abstract

Completion of Astra #9's R5. For the analytic family with common envelope (C_0, L, D, ρ) and every target T there is a constant $K_T = \text{uniformTreeConst}$, depending only on $(\beta, b, \rho, C_0, L, p_1, p_2, T, h, k, D)$, such that for EVERY member c of the family and EVERY $N \geq 1$,

$$\left| Z_c(N) - \sum_{\alpha \in \Lambda(h, k), \alpha < 2T} N^{-\alpha} (A_\alpha \log N + B_\alpha) \right| \leq K_T N^{-2T} (1 + \log N)$$

(twoD_taylor_tree_uniform). The constant is the uniform cutoff constant of unit 133 at the cutoffs of unit 129 plus the envelope bounds of unit 134 on the discarded pole terms. Zero sorry/axiom.

1 The things formalised

```
theorem pos_of_mem_poleSet (h h k k M M : ) (hk : 0 < k) (hk : 0 < k) ( : )
  (h : poleSet h h k k M M) : 0 <
noncomputable def coeffEnv (b C L : ) (h h k k D : ) ( : ) : :=
  canonCEnv C h h k k * envMoment 0 (gaussEnv L D)
  + (canonUEnv b C h h k k * envMoment 0 (gaussEnv L D)
    + canonVEnv b C h h k k * envMoment 0 (gaussEnv L D)
    + canonCEnv C h h k k * envMoment 1 (gaussEnv L D))
noncomputable def uniformTreeConst ( b C L p p T : ) (h h k k D : ) : :=
  uniformConst b C L h h k k (cutoffMg k p (cutoffQ p p T))
  (cutoffMg k p (cutoffQ p p T)) D
  + (poleSet ... at those cutoffs).filter (fun => ⌊ ⌋ < 2 * T),
  coeffEnv b C L h h k k D
theorem twoD_taylor_tree_uniform ( b p p C L : ) (h h k k D : ) (h : 0 < )
  (hb : 0 < b) (hb : b < ) (hk : 0 < k) (hk : 0 < k) (hp : ((h : ) + 1) / k = p)
  (hp : ((h : ) + 1) / k = p) (c : × → → ) (hcc : ij, Continuous (c ij))
  (H : → ) (hH : Continuous H) (hc : ij s, |c ij s| * ^ (ij.1 + ij.2) H s)
  (hC : 0 C) (henv : s, 0 s → H s C * (1 + s) ^ D * Real.exp ( * s * L))
  (TN : ) (hN : 1 N) :
|twoDAmp b N h h k k (anaAmp c b)
  - polesBelowGen h h k k p p T,
  N ^ (-) * (canonA h h k k (fun i j s => c (i, j) s) * Real.log N
    + canonB b h h k k (anaFaceU c b) (anaFaceV c b)
      (fun i j s => c (i, j) s) )|
uniformTreeConst b C L p p T h h k k D
  * (N ^ (-(2 * T)) * (1 + Real.log N))
```

2 Mathematics

$Z - \sum_{\alpha < 2T} = (Z - S^{\text{merged}}) + \sum_{\alpha \geq 2T} \text{retained}$, where the merged sum equals the canonical pole sum over the whole retained set (unit 129 regrouping). The first term is bounded by the uniform cutoff constant at exponent $E \geq 2T$, pushed to exponent $2T$ (unit 131's monotonicity); each discarded term is at most $(|A_\alpha| + |B_\alpha|)N^{-2T}(1 + \log N)$ and $|A_\alpha| + |B_\alpha|$ is bounded by the envelope constants of unit 134 (all retained poles are positive). Summing gives K_T .

3 Paper-facing meaning

This is the strongest form of `thm:TaylorTree` for $d = 2$ obtained so far: the expansion holds with an explicit remainder constant that is uniform over all chart amplitudes $\eta_w e^{\beta s \xi_w}$ whose coefficient arrays share a weighted bound, for every $N \geq 1$ (not only eventually). In the SLT application the noncritical coordinates w range over the support of a partition of unity, and ξ, η analytic on a common neighbourhood give exactly such a common bound; so the per-stratum integral $\int Z_w(N) \rho_I(w) dw$ inherits the expansion with coefficients $\int A_\alpha(w) \rho_I$, $\int B_\alpha(w) \rho_I$ once measurability in w is established (the remaining piece of the noncritical-coordinate adapter, Astra #6 rank 4).

4 Lean notes

- Wrapping many intermediate objects in `set` (cutoffs, coefficient functions, the pole set, the remainder scale R) keeps the long statement manageable; a final `show` restates the goal with `polesBelowGen` unfolded to `S.filter`.
- `bound_mono_exp` produces $K * N \wedge (-E) * (1 + \log N)$; regroup with `← mul_assoc` against the `set` variable R .
- The unit compiled on the first attempt.